

IAM (Identity and Access Management) in AWS

IAM is a security service in AWS that controls access to resources based on individual identities. It allows you to assign specific permissions to new users, such as colleagues, to control what they can read, delete, or create.

Real-World Analogy: Think of IAM access like inviting friends or relatives into your home. You decide who can enter specific rooms based on your trust in them or their needs. Similarly, in IAM, you grant access to AWS resources based on the role or work of each user.

Steps to Use IAM:

1. Go to the **IAM Dashboard**.
2. **Create a new user:**
 - Set up a username and password.
 - While creating the user, assign permissions based on their role or tasks.
3. After creating the user, the credentials will allow the user to log in with the specific access levels you've set.

Authentication vs. Authorization:

- **Authentication** is about creating the user (confirming identity).
- **Authorization** is about assigning the correct permissions (granting the right access to resources).

Real-World Example: In real life, authentication is like verifying someone's identity before letting them into your house. Authorization is like deciding which rooms they can enter once inside.

Assigning Permissions: To give permissions:

1. Go to **Users**, then **Permissions**.
2. Choose how to assign permissions:
 - Add the user to **groups** (recommended for better management).
 - Copy permissions from another user.
 - Attach a policy directly (e.g., giving full EC2 access).
3. You can also assign **groups** that hold a predefined set of permissions. Adding a user to a group will automatically give them the group's permissions.

Policies:

- A **policy** is a collection of permissions written in **JSON format**.
- Policies consist of:
 1. **Effect:** Allow or deny access.
 2. **Actions:** The actions a user can take (e.g., `ec2:StartInstances`).

3. **Resources:** The specific AWS resources affected.

AWS provides managed policies that you can attach to users or groups. However, if needed, you can create custom policies based on your requirements:

- Go to **Policies**, then **Create Policy** using JSON or the visual editor.

If you are unsure how to write JSON, AWS provides tools to help generate policies using a web interface. This is a key skill for DevOps engineers, who must be resourceful and use tools like the AWS Policy Generator or ChatGPT to create policies.

Roles:

- **Roles** in IAM allow you to assign permissions to AWS services, instances, or users from other AWS accounts or regions. For example, you can assign a role to an EC2 instance that allows it to access an S3 bucket.

Real-Time Example: Consider a server that needs to access an S3 bucket. Instead of creating user credentials, you can create a **role** with the necessary permissions, and then assign it to the server.

Creating a Role:

1. Go to **Roles** in the IAM dashboard.
2. Define the permissions and configuration required for the role.
3. After creating the role, you can assign it to an AWS service (e.g., EC2 or Lambda).

Commands: To check permissions, you can use the command `aws iam list-roles` to view existing roles. You can also update roles by attaching new policies or assigning them to services.